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Antisocial but inconsequential: large language model agents punish cooperators yet neither sustain cooperation nor reproduce its cross-cultural geography

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Costly peer punishment is a cornerstone human institution for sustaining cooperation, yet its effectiveness and its antisocial misuse vary sharply across societies. We ask whether large language model (LLM) agents reproduce this institution and its cultural structure by running three open instruction-tuned models (Qwen2.5-7B, Gemma-2-9B, Llama-3.1-8B) through Herrmann's public-goods game with peer punishment, crossed with five languages, four dispositions and sixteen neutral society personas drawn from the original participant cities. The agents contribute generously (baseline means 14.26, 13.41 and 10.67 of 20 tokens against a human grand-mean of 8.6) and they punish heavily, but a majority of all deduction points (54–59%) are antisocial, aimed at members who contributed at least as much as the punisher, rivalling the most punitive human pools. Two diagnostic human results nonetheless fail to emerge. Punishment does not raise cooperation ($P-N = -0.36, -3.55$ and $+0.08$; Gemma's drop $p < 0.001$; equivalence tests bound the other two effects below half the human gain) against a human gain of $+4.3$, and the cross-societal signature collapses: the human antisocial-punishment-cooperation coupling ($\rho \approx -0.90$) vanishes ($\rho = -0.09, -0.15, +0.30$) and the geography of cooperation disappears. LLM agents reproduce the existence of (anti)social punishment but neither its institutional function nor its cultural structure.

Among the most striking findings in the experimental study of human cooperation is that people will pay to punish free-riders, that this costly punishment can stabilise contributions to a public good, and that its character varies markedly across societies—in some participant pools punishment is turned not against shirkers but against generous cooperators, an antisocial impulse that tracks weak norms of civic cooperation. As large language model agents are increasingly proposed as proxies for human decision-makers, it is natural to ask whether they reproduce this institution: do they punish at all, do they punish the right targets, does punishment sustain their cooperation as it does ours, and does its cultural geography survive when the same model speaks as a participant from sixteen different cities? Here we put three open models through Herrmann's classic design to find out.

1. Introduction

The provision of public goods poses one of the most enduring problems in the social and biological sciences: rational individuals who share in a collective benefit have a private incentive to withhold their own contribution, so that universal free-riding can unravel an outcome from which all would gain [1, 2]. In the laboratory, this tension is distilled in the linear public-goods game, where players repeatedly decide how much of a private endowment to invest in a common project whose returns are shared regardless of investment; absent further structure, contributions decay towards the free-riding equilibrium over successive rounds [1]. A large body of work shows that this decay can be arrested when players are permitted to punish one another at a personal cost, sacrificing their own payoff to reduce that of low contributors: such costly peer punishment functions as a decentralised institution that sustains cooperation where material self-interest alone would erode it [3, 4], and a substantial evolutionary literature has since examined when costly punishment and related institutional designs can emerge and stabilise collective action [5–7]. Yet the form this institution takes is far from universal. In a sixteen-society comparison spanning participant pools from Boston to Athens to Riyadh, [8] documented striking cross-societal heterogeneity, including widespread *antisocial* punishment—deduction points aimed at cooperators who contributed as much as or more than the punisher—whose prevalence covaried negatively with the cooperation a society could ultimately achieve. The capacity of a punishment institution to support collective action thus depends not merely on its existence but on culturally structured norms governing who gets punished and why, in line with broader accounts of how communities sustain commons through locally tailored rules and sanctioning [9].

Recent advances in large language models (LLMs) have prompted a growing interest in using them as simulated subjects for the social and behavioural sciences. Because an instruction-tuned model can be conditioned on a persona and asked to respond to the same materials presented to human participants, researchers have proposed treating LLM outputs as “silicon samples” that may reproduce population-level patterns of attitude and behaviour [10]. Such samples have been used to emulate survey respondents and economic agents [11], to populate generative agent societies whose interactions exhibit emergent social dynamics [12], and to replicate classic findings from psychology and experimental economics with surprising fidelity [13]. At the same time, the behavioural validity of these simulations remains contested: models can match aggregate human tendencies on some tasks while diverging systematically on others, silicon-sample conclusions can hinge on defensible-but-arbitrary analytic choices [14], and the conditions under which silicon samples track—rather than merely caricature—human behaviour are still being mapped [15, 16]. Establishing where the correspondence holds and where it breaks is therefore a

precondition for any scientific use of LLM agents as stand-ins for human populations, and a central question for the emerging social physics of artificial intelligence [17].

A natural testing ground is the multi-agent strategic game, and a number of studies have begun to place LLMs in repeated and one-shot interactions to probe their cooperative and competitive tendencies [18–20]. This work has shown that models can be induced to cooperate, defect, and respond to opponents’ moves, and that their choices are sensitive to prompt framing, to the language of play and to the social roles they are assigned [21–23]; game-theoretic audits have likewise begun to map systematic cooperation biases and incentive sensitivity across models and languages [24, 25]. Sanctioning itself has recently begun to attract attention: reasoning-class models free-ride when allowed to choose between sanctioning and sanction-free institutions [26], punishment strategies shape the emergence of cooperative norms in LLM-agent societies and common-pool settings [27, 28], generative agent-based models can replicate landmark third-party-punishment experiments under persona prompting [29], and LLMs punish unfairness more severely than humans in third-party settings [30]. What remains untested, however, is whether LLM agents reproduce the deeper *institutional function* of a sanctioning mechanism rather than its surface behaviour—whether granting them the means to punish actually sustains cooperation as it does for humans—and whether they reproduce the culturally heterogeneous punishment norms, including the antisocial/prosocial decomposition of punishment, that distinguish one human society from another. These are precisely the diagnostic features that make peer punishment a substantive scientific phenomenon rather than a generic propensity to penalise, and they have gone largely untested across models, languages, and simulated societies—even though survey evidence already suggests that country personas only partially shift the culturally homogeneous default profiles of current models [31].

We address this gap with a faithful replication of [8] on the FAIRGAME multi-agent framework [32]. Groups of four agents play a ten-period public-goods game under partner matching, each period receiving a fresh endowment of twenty tokens to allocate between a private account and a group project with a marginal per-capita return of 0.4. We run two treatments: a baseline N treatment with contribution only, and a P treatment that adds a costly punishment stage in which each agent may assign zero to ten deduction points to each other member at a one-to-three cost ratio, with co-players relabelled every period to block identity tracking. The full design crosses these treatments with three open instruction-tuned models—Qwen2.5-7B-Instruct, Gemma-2-9B-it, and Llama-3.1-8B-Instruct [33–35]—five languages, four behavioural dispositions, and sixteen neutral society personas drawn from Herrmann’s participant cities, each stated as a bare geographic origin with no behavioural stereotype, for ten groups per condition.

Our central contribution is a clear dissociation between the existence of (anti)social punishment and its institutional function and cultural structure. The agents do contribute and do punish, and a majority of their deduction points—fifty-four to fifty-nine per cent across models—is antisocial, directed at co-players who contributed at least as much as the punisher, a share that rivals the most punitive human pools. But the two diagnostic human results fail to emerge. First, the punishment institution does not raise cooperation: the P-minus-N change in mean contribution is -0.36 for Qwen (not significant), -3.55 for Gemma (Mann–Whitney $p < 0.001$), and $+0.08$ for Llama (not significant), against a human grand-mean gain of $+4.3$, even though baseline contributions are high (N-treatment means of 14.26, 13.41, and 10.67 of twenty, versus a human grand mean of 8.6); the contribution-by-period and deviation-binned punishment patterns are summarised in figure 1 and table 1. Second, the cross-societal signature collapses: the strong human negative coupling between antisocial punishment and cooperation ($\rho \approx -0.90$) is absent in the models ($\rho = -0.09$, -0.15 , and $+0.30$), the “geography of cooperation” disappears as sixteen city personas shift mean contribution by only about one token, and LLM-versus-human cross-societal rank correlations are merely $+0.33$, $+0.06$, and $+0.17$ (figure 2). LLM agents thus reproduce the *existence* of (anti)social punishment but neither its institutional *function* of sustaining cooperation nor its cultural *structure*, which we take as the explicit validity criterion against which silicon-sample replications of human cooperation experiments should be judged.

2. Methods

(a) The public-goods game with punishment

We replicate the public-goods game with costly peer punishment of Herrmann et al. [8]. Four players are matched as partners for the whole game and play ten periods. Each period every player receives a fresh endowment of 20 tokens and contributes an integer amount between 0 and 20 to a group project with a marginal per-capita return of 0.4, so that each contributed token returns 0.4 tokens to all four members. In treatment N players only contribute. In treatment P the contribution stage is followed by a costly punishment stage in which each player may assign 0–10 deduction points to each other member; every point costs the punisher one token and removes three tokens from the target, the canonical 1:3 cost ratio. To block cross-period identity tracking, the other members are re-labelled every period. There is no communication, and we run ten independent groups per cell. Following Herrmann et al. [8], we classify punishment by the contribution deviation between target and punisher: punishment of a target who contributed equal to or more than the punisher (deviation ≥ 0) is *antisocial*, whereas punishment of a lower contributor (deviation < 0) is punishment of free-riding.

(b) Agents, models and protocol

Each agent is an instruction-tuned LLM that receives a single natural-language prompt comprising a brief of the game rules, the history of play so far, and a disposition or society block, after which it reasons briefly and emits a parseable decision line—CONTRIBUTION = X in the contribution stage and DEDUCT: A=a B=b C=c in the punishment stage. Within each stage, all four agents’ decisions are collected in a single lockstep batch so that play is genuinely simultaneous. The protocol is built on the FAIRGAME multi-agent framework [32]. We serve three open-weight instruction-tuned models, Qwen2.5-7B-Instruct [33], Gemma-2-9B-it [34] and Llama-3.1-8B-Instruct [35], via vLLM at temperature 0.8 with a fixed seed to make runs reproducible; all simulations were executed offline on a single RTX PRO 6000. Exact model identifiers, decoding settings and per-condition game counts are listed in the electronic supplementary material. Each game is crossed with five languages (English, French, Arabic, Chinese, Vietnamese) and four dispositional prompts (neutral, cooperative, selfish, vengeful); we additionally cross sixteen society personas drawn from the participant cities of Herrmann et al. [8] (Boston, Copenhagen, Zurich, St. Gallen, Nottingham, Bonn, Seoul, Melbourne, Chengdu, Minsk, Samara, Dnipropetrovsk, Muscat, Istanbul, Riyadh and Athens), each rendered as a neutral “You are a participant from <City>, <Country>.” persona that carries no behavioural stereotype.

(c) Measures and analysis

For each game we record the mean per-period contribution (of 20 tokens) and, in treatment P, the punishment assigned by each member. The antisocial share is the antisocial deduction points divided by all deduction points, and we additionally bin punishment expenditure by the contribution deviation between target and punisher. The punishment-institution effect is summarised as the contribution difference P–N and tested with the Mann–Whitney U test on the ten per-group means. All human benchmark values are the published summary statistics of Herrmann et al. [8] (per-pool mean contributions over the sixteen participant pools; grand means 8.6 and 12.9 tokens in the N and P treatments). Because two of the central results are nulls, we quantify evidence *for* them rather than relying on failure to reject: the P–N contrast is accompanied by a two-one-sided-tests (TOST) Welch equivalence test [36] whose smallest effect size of interest is half the human grand-mean gain (± 2.14 tokens) and by a JZS Bayes factor BF_{01} [37], and each model’s antisocial-punishment-cooperation coupling is tested directly against the human $\rho \approx -0.90$ with a Fieller-corrected Fisher- z test. The cross-societal signature of Herrmann et al. [8] is assessed with the Spearman rank correlation between per-society antisocial punishment and mean contribution, and the alignment of the model’s geography of cooperation with the human ordering across the sixteen societies is evaluated with

Spearman rank correlations and Wilcoxon signed-rank tests against the human pool means. We report bias-corrected effect sizes (Hedges g and the rank-biserial correlation for the P–N contrast) with 95% confidence intervals and Holm–Bonferroni-adjusted p -values; the full effect-size tables, the per-society breakdown and the dispositional analysis appear in the electronic supplementary material. Throughout, we log the `parse_fallback` (format non-compliance) rate for every cell and report it as a joint data-quality and fairness signal.

3. Results

In the contribution-only treatment (N) all three models sustained substantial cooperation, with baseline mean contributions of 14.26 ± 0.48 (Qwen), 13.41 ± 0.37 (Gemma) and 10.67 ± 0.45 (Llama) tokens of the 20-token endowment (table 1)—comfortably above the human grand-mean of 8.6 tokens [8]. The diagnostic test, however, is whether adding a costly peer-punishment stage (treatment P) lifts cooperation, the central result of the punishment literature [4, 8]. It does not. The within-model difference $P - N$ was -0.36 tokens for Qwen, -3.55 for Gemma and $+0.08$ for Llama, against a human grand-mean gain of approximately $+4.3$ tokens (from 8.6 under N to 12.9 under P). For two of the three models the punishment institution thus left cooperation flat or, in Gemma’s case, drove it significantly *down* (Mann–Whitney $p < 0.001$; the per-period $N > P$ trajectory is shown in figure 1a), while for Qwen the small decline was not significant ($p = 0.71$) and Llama’s near-zero change likewise did not differ from chance ($p = 0.91$). These two nulls are supported positively rather than by mere non-significance: TOST equivalence tests establish that the institutional effect for Qwen and Llama is reliably smaller than half the human gain (TOST $p = 0.010$ and 0.002), with JZS Bayes factors favouring no treatment difference ($BF_{01} = 2.3$ and 2.5), whereas for Gemma the same analyses decisively reject both the null and equivalence in the direction of a harmful institution ($BF_{01} = 1.8 \times 10^{-4}$, overwhelming evidence for a difference; TOST $p = 0.99$). The institutional responsiveness that makes punishment informative about human conditional cooperation is therefore absent.

The punishment that the models did assign, moreover, was substantially antisocial. Classifying each deduction by the contribution deviation between target and punisher, antisocial punishment—points aimed at a target who contributed equal to or more than the punisher (deviation ≥ 0)—accounted for 59%, 55% and 54% of all points for Qwen, Gemma and Llama respectively, rivalling the most punitive human participant pools [8]. Yet the deviation-binned structure of that punishment (figure 1b) reveals that the models differ sharply in whether they retain the human-like logic of punishing deeper free-riding more harshly. Only Gemma reproduced the monotone gradient, escalating from 2.45 points in the mild-free-riding bin to 3.71 points for the most extreme free-riders (deviation ≤ -11) while sparing equal contributors (1.40 points at deviation 0).

Qwen and Llama, by contrast, punished almost uniformly across deviation bins, meting out comparable deductions to free-riders and cooperators alike: Qwen ranged only from 2.15 to 2.78 points and Llama from 2.53 to 2.86 points across the full deviation spectrum, punishing equal-or-higher contributors (2.29 and 2.60 points at deviation 0) nearly as hard as the worst free-riders. The behavioural surface of punishment is present, but for two models the underlying targeting logic that distinguishes prosocial from antisocial sanctioning is not.

Most consequentially, the cross-societal signature that makes Herrmann’s study diagnostic of culturally heterogeneous norms fails to emerge. In the human data, pools where antisocial punishment is prevalent are those where cooperation is low, yielding a strong negative rank correlation of approximately -0.90 between per-society antisocial punishment and mean contribution [8]. Across our sixteen neutral city personas this relationship was effectively flat: the antisocial-versus-cooperation Spearman correlation was -0.09 for Qwen, -0.15 for Gemma and $+0.30$ for Llama (figure 2a)—in Llama’s case of the opposite sign. Each of these couplings differs significantly from the human $\rho \approx -0.90$ over the same sixteen pools (Fieller-corrected Fisher- z tests; $p = 0.0006$, 0.0011 and $p < 10^{-4}$, respectively). The accompanying “geography of cooperation” also collapsed: society personas shifted contribution by only about one token, compressing all sixteen cities into a narrow band rather than reproducing the broad human cross-cultural spread (figure 2b). The models did not even rank the societies as humans do, with LLM-versus-human cross-societal rank correlations of just $+0.33$ (Qwen), $+0.06$ (Gemma) and $+0.17$ (Llama), and the overall level of cooperation differed from the human pools for two of the three models (Wilcoxon $p = 0.005$ for Gemma and $p = 0.025$ for Llama, mean LLM–human differences of -3.30 and -2.47 tokens; Qwen did not differ, $p = 0.67$). Thus, even where the models do punish, contribute and engage in antisocial sanctioning, the culturally structured preference heterogeneity that is the scientific payload of the original experiment is missing.

4. Discussion

The public-goods game with peer punishment offers one of the most demanding behavioural benchmarks in experimental economics, because it couples a first-order cooperation decision with a second-order sanctioning decision and asks whether the latter can stabilise the former. Our results show that large language model agents clear the surface of this benchmark while failing its diagnostic interior. The agents contribute, and contribute generously: baseline mean contributions in the contribution-only treatment reach 14.26, 13.41 and 10.67 tokens of the 20-token endowment for Qwen, Gemma and Llama respectively, all comfortably above the human grand mean of 8.6. The agents also punish, and a clear majority of the deduction points they spend are directed at group members who contributed an amount equal to

Table 1. Mean contribution (of 20 tokens) in the faithful baseline (English prompt, neutral disposition) without (N) and with (P) a punishment stage, the punishment effect $\Delta = \bar{c}_P - \bar{c}_N$, and the antisocial share of all punishment (points aimed at members who contributed equal to or more than the punisher). The human reference is the 16-pool grand mean of Herrmann *et al.* [8], where punishment raised cooperation ($\Delta \approx +4.3$) and antisocial punishment, though present, was a minority of expenditure in most pools. p from a two-sided Mann–Whitney U test on the ten per-group means (P vs N).

Player	Contrib. N	Contrib. P	Δ (P–N)	p	Antisocial share
Human [8]	8.6	12.9	+4.3	—	minority*
Qwen2.5-7B	14.26 ± 0.48	13.90 ± 0.51	−0.36	0.71	59%
Gemma-2-9B	13.41 ± 0.37	9.86 ± 0.37	−3.55	<0.001	55%
Llama-3.1-8B	10.67 ± 0.45	10.75 ± 0.43	+0.08	0.91	54%

*In Herrmann *et al.* the antisocial share varied widely across societies but prosocial punishment of free-riding dominated in most Western pools.

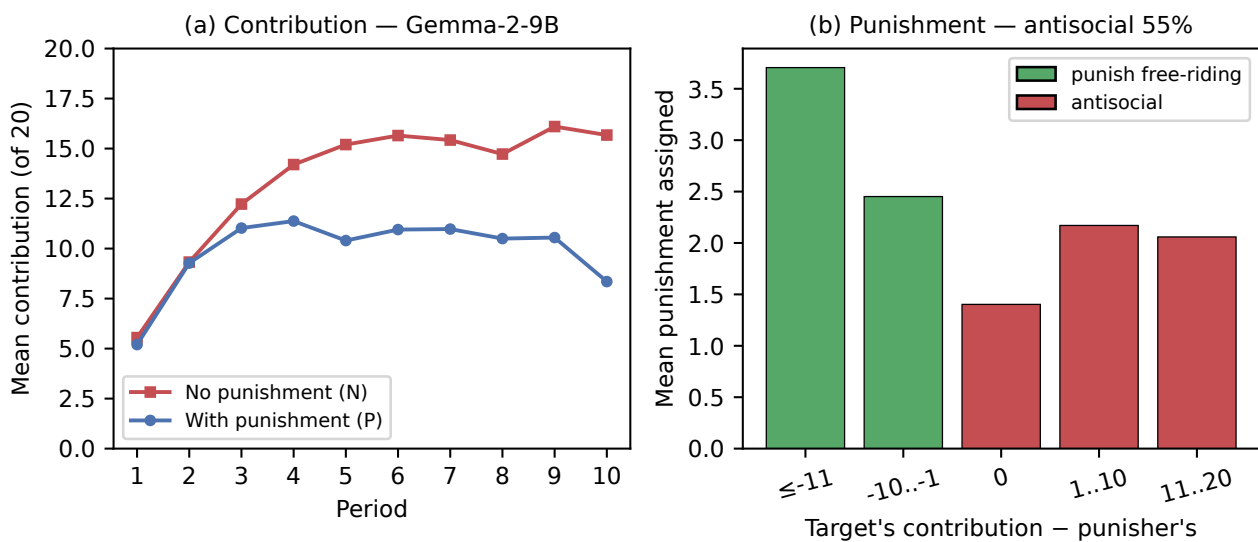


Figure 1. Punishment dynamics for Gemma-2-9B. (a) Mean contribution per period without (N) and with (P) punishment: the punishment treatment lies below the no-punishment baseline, the opposite of the human result that punishment sustains cooperation. (b) Mean punishment assigned by the contribution deviation of the target relative to the punisher; green bars are prosocial punishment of free-riding, red bars are antisocial punishment of equal or higher contributors, which accounts for 55% of all expenditure.

or greater than the punisher's own contribution. This antisocial share lies between 54% and 59% across the three models, placing every agent population at or beyond the antisocial-punishment extreme of the human cross-cultural distribution, rivalling the most punitive participant pools in Herrmann *et al.* [8]. In the narrow sense of behavioural repertoire, then, the agents reproduce the existence of both prosocial and antisocial punishment.

What they do not reproduce are the two results that make the human data theoretically load-bearing. The first is institutional. In human groups, the option to punish raises cooperation: across the sixteen societies of Herrmann *et al.* [8], opening the sanctioning stage lifts mean contributions by roughly 4.3 tokens, and in the canonical partner-matching design of Fehr and Gächter [3] punishment converts a decaying public good into a stable one. Our agents show no such institutional function. The contrast

between the punishment and no-punishment treatments, $P - N$, is -0.36 for Qwen and $+0.08$ for Llama, neither significantly different from zero, and -3.55 for Gemma, a significant *decrease* (Mann–Whitney $p < 0.001$; table 1, figure 1). The sanctioning institution that defines the human result is, for these agents, either inert or actively corrosive. Because so much of the punishment they deliver lands on equal and high contributors rather than on free-riders, the threat that should discipline defection instead penalises cooperation, and a cooperator weighing whether to give has no reason to expect that giving will be rewarded or that withholding will be punished. The behaviour is present; the conditional-cooperation logic that would let it do institutional work is absent.

The second failure is cultural. The most distinctive finding of Herrmann *et al.* [8] is not that antisocial punishment exists but that its prevalence varies

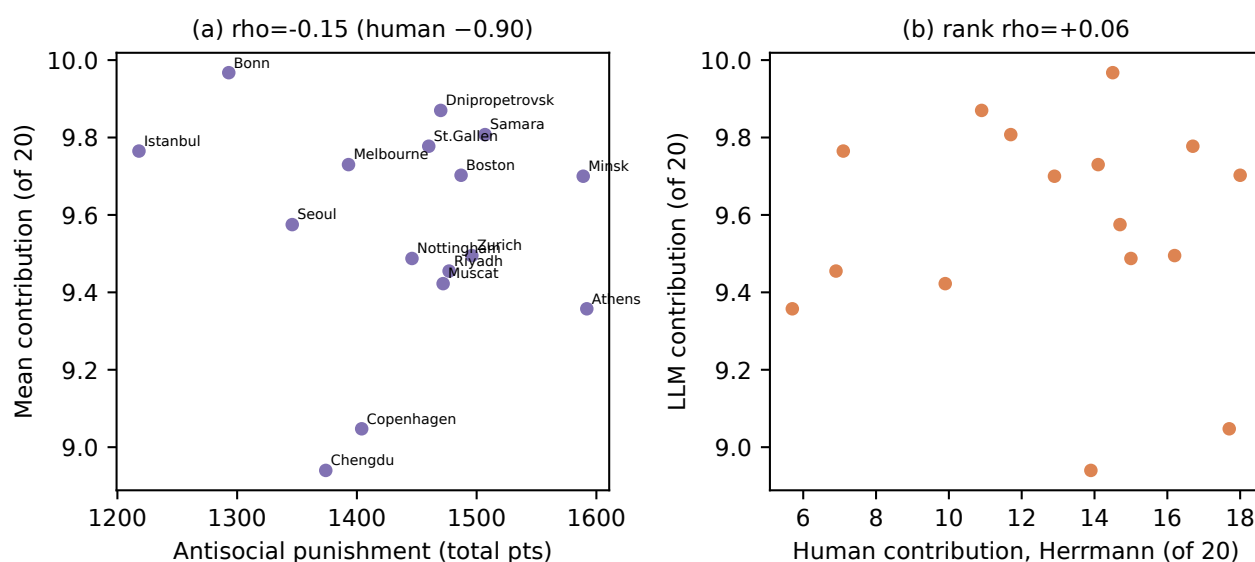


Figure 2. The cross-societal signature is not reproduced. (a) Per-society antisocial punishment versus mean contribution for the 16 city personas (Gemma-2-9B): the strong negative association seen in humans (Spearman $\rho \approx -0.90$) is absent ($\rho = -0.15$). (b) LLM versus human [8] mean contribution across the same societies: the model compresses every society into a ~ 1 -token band and does not reproduce the human ranking (rank $\rho = +0.06$).

systematically across societies and is tightly, negatively coupled to cooperation: where citizens punish high contributors, the public good collapses, yielding a cross-societal Spearman correlation of roughly -0.90 between antisocial punishment and contribution, and a recognisable “geography of cooperation” in which civic norms and rule of law track behaviour. None of this structure survives the transfer to silicon. Across our sixteen neutral city personas, the agent-level coupling between antisocial punishment and cooperation is effectively nil ($\rho = -0.09$, -0.15 and $+0.30$ for Qwen, Gemma and Llama; figure 2), and the geography itself flattens: changing the city named in the persona moves mean contribution by only about one token. This is consistent with survey-based evidence that cultural prompting only partially aligns LLM outputs with country-level human value profiles [31], although synthetic cultural agents have shown qualitative resemblance to non-WEIRD behaviour in simpler two-player fairness games [38]. When we rank the sixteen societies by agent cooperation and correlate those ranks against the human ordering, the agreement is negligible ($\rho = +0.33$, $+0.06$ and $+0.17$), and the absolute levels differ significantly for two of three models (Wilcoxon $p = 0.005$ for Gemma, $p = 0.025$ for Llama). The agents do not merely compress the human cross-cultural variance; they fail to reconstruct its ordering at all. The internal structure of punishment tells the same story. Only Gemma reproduces the human monotone gradient in which deeper free-riding is punished harder (3.71 points for the largest negative deviations versus 2.45 for moderate ones, while sparing equal contributors at 1.40 points); Qwen and Llama punish almost flatly across deviation bins (roughly 2.15–2.78 and 2.53–2.86

points), hitting cooperators nearly as hard as defectors. The deviation-sensitivity that gives human punishment its corrective meaning is, for two of three models, simply not there.

Taken together, these findings draw a sharp line between reproducing a behaviour and reproducing its function. It is increasingly common to treat LLM agents as “silicon subjects” whose responses can stand in for human participants in survey and economic experiments [10–13]. Our results sound a specific caution for that programme. The agents pass the descriptive test—they produce contribution and punishment distributions that, viewed marginally, look broadly human and even capture the surprising prevalence of antisocial sanctioning. Yet the two relational properties that any explanatory use of the paradigm depends upon—that punishment causally sustains cooperation, and that its antisocial form varies across cultures and predicts cooperative collapse—are exactly the properties that vanish. A silicon-subject study that read off only the marginals would conclude, wrongly, that these agents recover the Herrmann result; the failure is visible only in the conditional and cross-societal structure. Behavioural realism at the level of distributions is therefore a weak guarantee of mechanistic realism, and the use of LLM agents to forecast institutional outcomes should be validated against precisely the diagnostic, relational signatures that distinguish competing theories—in line with recent statistical frameworks for validating LLM simulations as behavioural evidence [16] and with warnings that silicon-sample results can hinge on analytic flexibility [14]. A companion study reaches the same

conclusion for catastrophe-risk sensitivity in the collective-risk climate dilemma, where the same models reproduce high cooperation but not its dependence on risk [39].

The implications extend beyond the methodology of computational social science to the engineering of multi-agent LLM systems. There is growing interest in populations of language agents that negotiate, divide resources and self-govern [12, 18, 21, 28, 40–42], and costly peer punishment is one of the few decentralised institutions known to stabilise cooperation among self-interested humans without a central authority. Our results indicate that this institution does not transfer for free. When the same sanctioning channel that is supposed to discipline defectors is instead used, the majority of the time, against good citizens—and when, for one model, equipping agents with punishment lowers the collective payoff relative to no punishment at all—the mechanism that designers might reach for to keep an agent society cooperative becomes a liability rather than a safeguard. A multi-agent system that grants its members the ability to sanction one another cannot assume that the ability will be exercised correctively; absent the conditional-cooperation and deviation-tracking logic that humans bring to the task, sanctioning power may amplify rather than damp conflict, turning the punishment institution into a tax on cooperation—a concrete instance of the broader lesson that mechanisms promoting cooperation need not promote social welfare [43].

(a) Limitations

Several limitations bound the scope of these conclusions. First, we study three open-weight instruction-tuned models in the 7–9B parameter range (Qwen2.5-7B-Instruct, Gemma-2-9B-it, Llama-3.1-8B-Instruct; [33–35]) at a single decoding temperature of 0.8. We cannot say how much of the diagnostic failure reflects model scale or sampling, and it remains possible that larger frontier systems would recover more of the institutional and cultural structure; our claims are therefore claims about these models under these settings, not about language agents in general. Second, our societal manipulation is deliberately minimal: each persona names only a city and country from Herrmann’s participant pool, with no behavioural stereotype attached. This neutrality is a design choice that lets us ask whether a bare geographic label, of the kind a silicon-subject study might casually deploy, suffices to elicit cultural variation—and our null result speaks directly to that question. But it also means we have not tested whether richer, norm-laden personas could recover the geography of cooperation, and the absence of cross-societal heterogeneity should be read as a statement about neutral labels rather than about the models’ ceiling for cultural differentiation. Third, format non-compliance is negligible in the English baseline that carries the headline results (at most 1.8% of contribution decisions and 6.8% of punishment decisions per model) but rises sharply in some low-resource pairings, reaching 31% of contribution and 43% of punishment decisions for Gemma in Arabic and 15–19% of contribution decisions for

Llama in Vietnamese; although we log every cell’s rate as a joint data-quality and fairness signal, the affected cross-language cells carry correspondingly fewer clean decisions, and non-compliance may interact unevenly with personas and dispositions. Fourth, although the same four agents stay together for all ten periods, the design—faithful to the original—blocks communication, explicit agreements and cross-period identity tracking (co-players are re-labelled every period), so the rich reciprocity and reputational dynamics that scaffold human punishment are unavailable by construction. Fifth, like all prompted-LLM studies our results carry some prompt sensitivity, and the natural-language framing of the brief, history and disposition block could shift behaviour in ways we have not exhaustively mapped. Finally, instruction-tuned models are shaped by alignment training that explicitly discourages harmful or punitive outputs and encourages even-handedness, which may simultaneously inflate baseline contributions, blunt the deviation-sensitivity of punishment, and homogenise behaviour across personas; disentangling alignment effects from genuine strategic reasoning is beyond what our design can resolve.

These limitations chart the natural next steps. Personas richer than a neutral city label—carrying explicit social norms, institutional context or cultural values—would test whether the geography of cooperation is genuinely unreachable or merely under-elicited by bare geographic framing. Allowing communication, explicit agreements and cross-period reciprocity would let agents build the reputational scaffolding on which human punishment depends, and would reveal whether the institutional function of sanctioning emerges once that scaffolding is in place. Agents whose policies are shaped by reinforcement learning in the game itself, rather than by prompting alone, would separate what current models do by default from what language agents can be trained to do. And extending the battery to frontier models would establish whether the diagnostic failures we report shrink with scale or persist as a structural feature of LLM agents.

5. Conclusion

In Herrmann’s public-goods game with costly peer punishment, large language model agents reproduce the existence of (anti)social punishment but neither its institutional function nor its cultural structure. They contribute generously and they punish, and a majority of their punishment—54% to 59% of all deduction points—is antisocial, aimed at members who gave as much as or more than the punisher, rivalling the most punitive human pools. Yet the punishment institution does not raise cooperation, and for one model significantly lowers it; the strong cross-societal coupling between antisocial punishment and cooperation collapses from a human $\rho \approx -0.90$ to near zero; and the geography of cooperation flattens, with sixteen neutral personas moving contribution by about a single token. The agents reproduce the behaviour while losing the relational logic that makes it matter.

For silicon-subject research this means that human-looking marginals are no guarantee of human-like mechanism, and that the diagnostic, conditional structure of a paradigm is where validation must focus. For multi-agent LLM systems

it means that the sanctioning institutions humans use to stabilise cooperation cannot be assumed to transfer—granted the power to punish, today's agents are as likely to turn it on good citizens as on free-riders.

Ethics. This study involved no human or animal participants. All behavioural data were generated by computational simulations of large language models and therefore required no ethics approval.

Data Accessibility. All materials needed to reproduce the study—the FAIRGAME-based simulation code, the prompt templates in five languages, the configuration files, the per-game result files for all three models, and the analysis script (`analysis/analyze.py`) that regenerates every figure, table and number—together with the electronic supplementary material are available in the project repository and will be deposited with a permanent DOI (Zenodo) on acceptance [DOI to be inserted]. The human benchmark is taken from the published data of Herrmann *et al.* [8].

Authors' Contributions. Author 1: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, and writing (original draft, review and editing). The author gave final approval for publication and agrees to be held accountable for the work performed therein.

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Disclaimer. Large language models were used solely as the objects of study (the simulated agents); all analysis, figures and text were produced and verified by the authors.

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